

A05_TEAM 3_MICHAEL_RIVAS_ITAI 2376

Arrival & NLP: Decoding Alien and Human Language

Team 3 Collaborative Analysis

Cherluk, Michael, Olu, & Taki

Who contributed what — and where the analysis deepens

CHERLUK

Plot framing and first-contact context. Extends the film overview into a linguistic fieldwork problem: vocabulary must be established before reliable interpretation.

OLU

Plot structure and multimodal communication. Extends the spoken-versus-logographic contrast into a comparison of sequential speech and simultaneous visual meaning.

MICHAEL

Lexical and semantic ambiguity. Extends the “weapon/tool” example into word-sense disambiguation, context dependence, and translation risk.

TAKI

Pragmatics, idioms, sarcasm, and cultural context. Extends the challenge beyond literal translation to intended meaning and social interpretation.

Shared responsibility: algorithmic mapping, synthesis, references, and rubric coverage connect the four individual analyses into one argument.

Analyzing linguistic barriers across species and systems.



Shared Analysis

Each member analyzed "Arrival" (2016) to evaluate extraterrestrial communication hurdles.



Real-World Parallels

Jointly linked film complexities to NLP challenges (ambiguity, idioms, culture).



Synthesized Insights

Combined distinct perspectives to critique modern NLP through an algorithmic lens.

From individual observations to a defensible NLP argument

01 OBSERVE

Each member identifies a communication problem in Arrival rather than treating the movie only as a plot summary.

02 TRANSLATE

The film observation is converted into an NLP concept: ambiguity, pragmatics, context, semantics, or representation.

03 TEST THE PARALLEL

The team asks what a real NLP system would need to resolve the same problem and where the analogy breaks down.

04 SYNTHESIZE

The final claim separates what the movie illustrates from what modern NLP can actually support.

This workflow directly strengthens the rubric areas for analysis, reflection, clarity, and collaboration.

PLOT OVERVIEW (CHERLUK & OLU)

Language at the Center of First Contact

- **Global Arrival:** Twelve vessels mandate urgent interspecies communication.
- **Dual Systems:** Spoken audio (A) versus circular logograms (B).
- **Linguistic Fieldwork:** Eliciting vocabulary to build shared lexicons.
- **Non-Linear Time:** Written symbols reshape human temporal perception.



Plot overview expanded into a linguistic case study

CHERLUK — FIELDWORK

The opening contact problem is not simply “translation.” It is vocabulary discovery under uncertainty: humans must determine what recurring signals refer to before interpreting more complex messages.

OLU — REPRESENTATION

The film presents two communication channels: spoken audio and circular written symbols. That distinction creates a useful bridge to speech processing, text processing, and multimodal NLP.

JOINT EXTENSION

The heptapod writing system challenges the assumption that meaning must arrive as a left-to-right sequence. Representation itself becomes part of the linguistic problem.

ASSIGNMENT VALUE

This section supplies the narrative foundation needed before evaluating specific NLP failures; it prevents the presentation from becoming a list of disconnected movie scenes.

Rubric connection: strengthens NLP Challenge Analysis, Clarity & Organization, and Creative Parallel Drawing.



NLP CHALLENGE 01 (MICHAEL)

Lexical & Semantic Ambiguity

The "Weapon vs. Tool" Crisis

At [01:14:00], the translation "offer weapon" nearly triggers global war, ignoring the benign semantic sense of "tool."

- **Meaning Compression:** Logograms encode simultaneous multiple meanings.
- **Polysemy Hurdles:** Single symbols map to several contextual senses.
- **Algorithmic Parallel:** Modern systems struggle with word-sense disambiguation.

Ambiguity: from “weapon” to word-sense disambiguation

OBSERVATION

The disputed “weapon” translation demonstrates how a single lexical choice can carry very different consequences when context is incomplete or competing interpretations are plausible.

NLP PARALLEL

Word-sense disambiguation attempts to select the contextually appropriate meaning of an ambiguous word or phrase. A system cannot safely rely on the surface form alone.

DEEPER ISSUE

The film also compresses several meanings into a non-linear symbol. That resembles a representation problem: the system must preserve relationships among concepts instead of treating meaning as one fixed label.

MICHAEL'S EXTENSION

The key contribution is to distinguish lexical ambiguity from translation accuracy: a grammatically plausible output can still be semantically unsafe when context is under-modeled.

Rubric connection: adds depth to the 20-point NLP Challenges category and creates a concrete bridge to real-world semantic processing.

Idioms, Sarcasm & Cultural Context



Pragmatics Failure

Literal mistranslation ignores intended conversational idioms.



Missing Audio Cues

Absence of vocal inflection obscures tone detection logic.



Cultural Filtration

Global powers interpret alien messages via Mahjong cultural logic.

Real-world impact: Idiom translation and sarcasm detection remain largely unsolved NLP problems without rich domain context.

Pragmatics: meaning is more than the literal words

IDIOMS

Idiomatic language can encode a meaning that cannot be recovered by translating each word independently. NLP therefore needs phrase-level and context-sensitive interpretation.

SARCASM

Sarcasm creates a mismatch between literal wording and intended meaning. Text-only systems may miss vocal tone, shared knowledge, or the situation that signals the intended interpretation.

CULTURAL CONTEXT

The same utterance can be interpreted differently across communities. The film dramatizes this through competing human assumptions about the visitors' symbols and intentions.

TAKI'S EXTENSION

The contribution separates semantics from pragmatics: knowing what a sentence can mean is not identical to knowing what a speaker intends by using it in a situation.

Rubric connection: strengthens the real-world NLP parallel and the assignment's requirement to address idioms, sarcasm, ambiguity, and cultural variation.

Decoding Methods vs. NLP Evolution

01

Rule-Based NLP

Hand-built formal grammars generated via whiteboard symbol-trials.

02

Statistical NLP

Probability-based inference relying on iterative pattern recognition.

03

Deep NLP

Contextual embeddings learned entirely through holistic language immersion.

600 × 400

Algorithmic approaches — what each member adds to the comparison

CHERLUK

Rule-based perspective: explicit vocabulary and grammar rules resemble the early stage of establishing stable mappings between symbols and concepts.

OLU

Representation perspective: the shift between speech and circular writing highlights why the input format matters before any statistical or neural model can interpret it.

MICHAEL

Statistical/deep perspective: contextual evidence is needed to resolve competing lexical meanings instead of choosing a translation from a fixed dictionary entry.

TAKI

Pragmatic perspective: context, social cues, and cultural knowledge expose limits that cannot be solved by literal token matching alone.

Joint conclusion: Arrival can be mapped onto the evolution from rules → probabilistic inference → contextual/deep learning, while also showing why context remains difficult.

Reflections on Meaning & Computation

→ The Sapir–Whorf Hypothesis

Language actively shapes cognition rather than merely expressing it.

→ Understanding vs. Prediction

LLMs predict token probabilities without genuine semantic comprehension.

→ Non-Linear Processing

Heptapod scripts severely challenge left-to-right sequential token assumptions.



Team Critique: The NLP Overreach

The film implies perfect semantic transfer is inevitable. We argue modern NLP lacks true grounded physical context to achieve this.

Four perspectives, one central question: what counts as understanding?

CHERLUK + OLU

The film treats language as a system that can reorganize how experience is represented. Their plot analysis makes representation a prerequisite for interpretation.

MICHAEL

Ambiguity shows why prediction of a plausible word is not the same as proving that the intended sense has been recovered.

TAKI

Pragmatics shows why meaning depends on speakers, situations, cultural expectations, and implied intent—not only on the text string.

TEAM SYNTHESIS

The presentation therefore distinguishes pattern recognition from grounded understanding. This makes the movie's non-linear language a prompt for evaluating assumptions behind modern NLP.

Reflective claim: the movie expands the NLP discussion from “Can a system translate?” to “What evidence would demonstrate that the system understood the intended meaning?”

References

Villeneuve, D. (Director). (2016). *Arrival* [Film]. Paramount Pictures.

Chiang, T. (1998). Story of Your Life. In *Stories of Your Life and Others*.

Sapir, E. (1929). The status of linguistics as a science. *Language*, 5(4), 207–214.

Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. *ACL 2020*.

Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing* (3rd ed. draft).

ITAI 2376 Course Staff. (2024). *Module 4-5 Slides: Machine Translation and Semantics*.

How the sources support the team analysis

FILM + FICTION

Villeneuve's *Arrival* and Chiang's *Story of Your Life* establish the narrative and speculative language concepts being analyzed.

LINGUISTICS

Sapir provides the historical linguistic framing used for the discussion of language, cognition, and the Sapir–Whorf hypothesis.

NLP RESEARCH

Bender & Koller and Jurafsky & Martin provide the academic bridge from fictional communication problems to questions of meaning, representation, and language processing.

COURSE CONNECTION

ITAI 2376 course materials anchor the discussion in the class's treatment of machine translation and semantics rather than relying only on outside sources.

Research contribution is distributed across the team analysis: each member's section uses the shared source base to connect a film observation to an academic NLP concept.

Thank You

Questions?

Team 3 — ITAI 2376

Cherluk, Michael, Olu, & Taki

Coverage checkpoint — every rubric category is visible

60 PTS — CONTENT

NLP challenges: ambiguity, idioms, sarcasm, culture. Communication methods: fieldwork, symbols, context, and algorithmic parallels. Reflection: meaning, cognition, and the limits of prediction. Organization: progression from plot → challenges → methods → synthesis.

20 PTS — RESEARCH

Film, fiction, linguistics, NLP scholarship, and course materials are identified. The bibliography remains visible as the evidence base for the presentation.

10 PTS — CREATIVITY

The “weapon/tool” case, non-linear writing system, and semantics-versus-pragmatics distinction create original parallels between the film and NLP.

10 PTS — TEAM

Named contribution areas make ownership visible: Cherluk + Olu for plot/representation, Michael for ambiguity, Taki for pragmatics, and all members for algorithmic synthesis and research.

Presentation-level takeaway: the added slides make individual work auditable while adding analysis rather than repeating the original slide text.

Image Sources



https://videocdn.cdnpk.net/videos/c362f365-24d5-56ef-9a2f-c7cab534364f/horizontal/thumbnails/large.jpg?sem=ais_hybrid&item_id=7860487&w=740&q=80

Source: www.magnific.com



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